THE TOPOLOGICAL SPACE OF A QUASI-COMPACT QUASI-SEPARATED ALGEBRAIC SPACE IS SPECTRAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 08YG: sources/stacks-chow/tags.txt, source lines 11339–11339. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 06NJ: sources/stacks-chow/tags.txt, source lines 8538–8538. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03E4: sources/stacks-chow/tags.txt, source lines 4443–4443. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03H6: sources/stacks-chow/tags.txt, source lines 4555–4555. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03BZ: sources/stacks-chow/tags.txt, source lines 4366–4366. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03H4: sources/stacks-chow/tags.txt, source lines 4553–4553. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AHR: sources/stacks-chow/tags.txt, source lines 13239–13239. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. The topological space $|X|$ is a spectral space.*

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Proof. By Topology, Definition 08YG we have to check that $|X|$ is sober, quasi-compact, has a basis of quasi-compact opens, and the intersection of any two quasi-compact opens is quasi-compact. By Lemma 06NJ we see that $|X|$ is sober. By Lemma 03E4 we see that $|X|$ is quasi-compact. By Lemma 03H6 there exists an affine scheme U and a surjective étale morphism $f : U \to X$. Since $|f| : |U| \to |X|$ is open and continuous and since $|U|$ has a basis of quasi-compact opens, we conclude that $|X|$ has a basis of quasi-compact opens. Finally, suppose that $A, B \subset |X|$ are quasi-compact open. Then $A = |X'|$ and $B = |X''|$ for some open subspaces $X', X'' \subset X$ (Lemma 03BZ) and we can choose affine schemes V and W and surjective étale morphisms $V \to X'$ and $W \to X''$ (Lemma 03H6). Then $A \cap B$ is the image of $|V \times_X W| \to |X|$ (Lemma 03H4). Since $V \times_X W$ is quasi-compact as X is quasi-separated (Lemma 0AHR) we conclude that $A \cap B$ is quasi-compact and the proof is finished. $\square$